PROJECT AGRENZA (FARMLINK)

Complete Cloud Deployment Guide & Architectural Reference

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SYSTEM PRODUCTION ARCHITECTURE SUMMARY:
- **Global Edge Frontend:** Next.js 14 App Router on Vercel CDN (*https://agrenza.vercel.app*)
- **Containerized Backend:** Apache PHP 8.2 REST API on Render.com (*https://agrenza.onrender.com*)
- **Distributed Database:** TiDB Cloud Serverless MySQL 8.0 Wire-Compatible Cluster
- **Authentication & Mailer:** Dual SSL Gmail SMTP on Port 465 & Firebase Auth Infrastructure
- **Source Control & CI/CD:** GitHub Auto-Deployment Pipeline (*Manikandan79-glitch/Agrenza*)

Tier Component	Technology	Provider	Live Production Endpoint / Host
Frontend UI	Next.js 14 (React 18 + TS)	Vercel	https://agrenza.vercel.app
Backend REST API	PHP 8.2 + Apache Docker	Render.com	https://agrenza.onrender.com
Relational DB	MySQL 8.0 Protocol	TiDB Cloud	gateway01.ap-northeast-1.prod.aws.tidbcloud.com:4000
Code Repository	Git Version Control	GitHub	github.com/Manikandan79-glitch/Agrenza
Transactional Mail	SMTP SSL Port 465	Google Gmail	smtp.gmail.com (ravananm913@gmail.com)

1. Executive Summary & Multi-Tier Architecture

Agrenza (formerly FarmLink) is a farm-to-doorstep direct agricultural marketplace application. The system replaces exploitative agricultural intermediaries with a transparent, direct transaction pipeline. To ensure continuous availability, high performance, and zero hosting costs during academic presentation, the application adopts an enterprise-grade **Decoupled Headless Multi-Tier Cloud Architecture**.

Key Architectural Advantages:

- **Headless Decoupling:** The user-facing presentation tier runs independently of the backend API, allowing edge-optimized caching and instant mobile page delivery.
- **Unified Relational State:** A centralized serverless MySQL instance guarantees real-time inventory consistency, farmer payouts, and order synchronization.
- **Continuous Integration / Continuous Delivery (CI/CD):** Every git push to GitHub automatically triggers automated builds on both Vercel and Render.
- **Cost Efficiency:** Implemented utilizing 100% free-tier cloud resources without sacrificing SSL encryption, multi-region routing, or database scalability.

2. Tier 1: Cloud MySQL Database (TiDB Cloud Serverless)

Relational data persistence was migrated from a local WAMP MySQL 8.4 engine to **TiDB Cloud Serverless**, an enterprise-grade distributed SQL database supporting native MySQL 8.0 wire compatibility and SSL/TLS encryption.

Parameter	Production Value	Description
Host / Endpoint	gateway01.ap-northeast-1.prod.aws.tidbcloud.com	AWS Tokyo High-Availability Cluster Gateway
Port	4000	Dedicated TiDB MySQL connection port
Database Name	farmlink	Production database catalog
User	8nQ2NS6oZwxnCRL.root	Cloud administrator username
SSL Mode	REQUIRED (MYSQLI_CLIENT_SSL)	Mandatory TLS in-transit encryption
Storage Quota	5 GiB Row-Based Storage (Free)	Permanent academic free tier allocation

The Seven Core Tables in the Schema:

Table Name	Primary Key	Key Foreign Keys	Purpose / Business Logic
users	id (INT)	None	Stores Admin, Farmers, and Buyers with roles, bcrypt passwords & status.
categories	id (INT)	None	Product categories (Vegetables, Fruits, Herbs, Spices, Dairy).
products	id (INT)	farmer_id, category_id	Item listings with base_price, GST, commission, stock_kg & approval.
orders	id (INT)	buyer_id -> users(id)	Customer orders, delivery address, payment mode, status & total.
order_items	id (INT)	order_id, product_id, farmer_id	Line-item breakdown tracking farmer payouts and platform cuts.
bulk_requests	id (INT)	buyer_id, product_id	Wholesale bulk purchase requests (15kg+) requiring admin review.
pending_registrations	id (INT)	None	Staging table holding unverified registrations until OTP validation.

Database Migration & Import Procedure:

1. Local database was exported using mysqldump -u root --default-character-set=utf8mb4 farmlink.
2. The cloud database catalog farmlink was provisioned via MySQL CLI.
3. The complete SQL dump (including DDL statements, constraints, and seed data) was executed directly over SSL.
4. Live integrity checks confirmed all 7 tables and 10 production product records active in cloud.

3. Tier 2: Backend REST API on Render.com

The PHP API backend was packaged into an immutable Docker container and deployed to **Render.com Web Services**. The container executes on a Debian Linux base with Apache 2.4 and PHP 8.2.

Production Dockerfile Specification:

```
FROM php:8.2-apache
   RUN docker-php-ext-install mysqli pdo pdo_mysql
   RUN a2enmod headers rewrite
   COPY farmlink/ /var/www/html/farmlink/
   WORKDIR /var/www/html/farmlink
   EXPOSE 80
```

Critical Architectural Fixes Applied in Backend:

• **Cloud SSL Driver Integration:** Updated farmlink/backend/config/db.php with mysqli_init() and mysqli_ssl_set() to allow secure handshakes with TiDB Cloud on port 4000.

• **Cross-Origin Resource Sharing (CORS):** Enhanced _bootstrap.php to detect request origins and issue dynamic Access-Control-Allow-Origin: https://agrenza.vercel.app headers with Access-Control-Allow-Credentials: true.

• **Cross-Domain Session Cookies:** Configured PHP session cookies with SameSite=None; Secure to ensure authentication tokens persist between Vercel and Render.

• **Dynamic Image Asset URL:** Implemented dynamic hostname and protocol resolution in UPLOAD_URL so product images render correctly across both localhost and agrenza.onrender.com.

Verified Live API Endpoints:

Endpoint URL	HTTP Method	Payload / Output	Status
/farmlink/backend/api/products/list.php	GET	JSON array of approved products with GST calculations	200 OK
/farmlink/backend/api/auth/login.php	POST	Body: { role, email, password } -> Session Cookie	200 OK
/farmlink/backend/api/auth/register.php	POST	Body: { role, name, mobile, email } -> Sends Gmail OTP	200 OK
/farmlink/backend/api/admin/farmers.php	GET	Admin query returning all approved & pending farmers	200 OK
/farmlink/backend/api/admin/buyers.php	GET	Dedicated endpoint returning customer details and spending	200 OK

4. Tier 3: Frontend Deployment on Vercel Edge Network

The Next.js 14 client application is deployed onto **Vercel's Global Edge Network**. The build process compiles React Server Components, client interactivity bundles, and dynamic routes into static assets and Edge Serverless functions.

Deployment Configuration Steps:

1. **GitHub Import:** Connected repository Manikandan79-glitch/Agrenza to Vercel via authorized GitHub App integration.
2. **Root Directory Alignment:** Explicitly configured the project Root Directory to farmlink-nextjs (since the repository contains both frontend and backend subdirectories).
3. **Build Pre-Verification:** Resolved strict TypeScript compilation constraints in bulk-request/page.tsx ensuring zero type-check errors during npx tsc --noEmit.
4. **API Base URL Routing:** Configured app/lib/api.ts with production fallback pointing directly to https://agrenza.onrender.com/farmlink/backend/api.
5. **Automatic CI/CD Pipeline:** Every commit pushed to the main branch immediately initiates an automatic Vercel production build.

5. Live Features & Mobile Verification Guide

The application has been verified for complete mobile and desktop responsiveness. All features interact with the live cloud database over HTTPS.

Feature / Route	Target Role	Functionality in Production
/shop	Public / Buyer	Real-time vegetable & fruit catalogue with GST-inclusive pricing and stock count.
/buyer/checkout	Buyer	Locked to Kanyakumari district; accepts only pincodes 629001 and 629401.
/admin/farmers	Admin	Displays all approved and pending farmers, Aadhaar records, and total sales.
/admin/buyers	Admin	NEW: Comprehensive customer table showing phone, email, orders count, and spent.
/admin/orders	Admin	Order management workflow with real-time status transitions (Pending -> Delivered).
/buyer/bulk-request	Buyer	Dynamic wholesale request form for quantities >= 15kg with live price calculator.

Mobile Testing Credentials:

- **Super Administrator:** URL: https://agrenza.vercel.app/admin/admin-login | User: admin@farmlink.com | Password: admin123
- **Farmer Portal:** User: sathishm250406@gmail.com | Password: FdXxs#43 | Access: /farmer/dashboard
- **Buyer Portal:** Self-register at /auth/buyer-register to test real-time Gmail OTP dispatch directly to your smartphone.

6. Frequently Asked Viva Questions & Model Answers

Q1: Why did you decouple Next.js and PHP into separate cloud hosts instead of using a single server?
Answer: Next.js performs optimal server-side rendering and static page generation on Vercel's global CDN edge nodes, resulting in sub-second page loads globally. Conversely, our existing backend leverages PHP PDO and Apache modules for MySQL interactions and mail dispatch. Decoupling allows each tier to operate on its specialized runtime without complex hybrid server configurations.

Q2: How do you handle Cross-Origin Resource Sharing (CORS) and session persistence between Vercel and Render?
Answer: The frontend (agrenza.vercel.app) and backend (agrenza.onrender.com) reside on distinct origins. In _bootstrap.php, we dynamically match the origin header against allowed patterns and respond with Access-Control-Allow-Origin and Access-Control-Allow-Credentials: true. Crucially, session cookies are configured with SameSite=None; Secure to ensure

modern mobile browsers transmit authentication cookies across distinct domains.

Q3: How does the system guarantee database security when connecting to TiDB Cloud over the public internet?
Answer: Plaintext database traffic is strictly forbidden. The connection invokes mysqli_ssl_set() with the MYSQLI_CLIENT_SSL flag, enforcing TLS 1.3 encryption for all SQL queries and result sets in transit between Render.com and the AWS Tokyo cluster gateway.

DECLARATION: This documentation reflects the true production architecture of the Agrenza Agricultural Marketplace system as implemented for the BCA degree evaluation at Government Arts and Science College, Palkulam. All services are currently live and verifiable via any web browser.